higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

ELECTROTECHNICS N4

28 MARCH 2019

This marking guideline consists of 7 pages.

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QUESTION 1

1.1 E
1.2 J
1.3 A
1.4 F
1.5 K
1.6 B
1.7 G
1.8 C
1.9 H
1.10 D

(10 × 1) **[10]**

QUESTION 2

2.1
- They heat up the transformers.
- They create a loss in the circuit.
- The power factor deteriorates.
- They are called eddy currents. (4)

2.2
- Hard magnet material✓
  Retain the magnetism and become permanent magnets. Used in meters telephone recievers✓

- Soft magnetic material✓
  Lose magnetism quickly (temporary). Used in generators and motors✓ (4)

2.3
- Separately excited
- Self-excited (2)

2.4 Increase in the temperature of material causes the resistance to rise (2)

2.5 2.5.1 I = 2 A

$$R_{//} = \frac{(R_1 + R_2)}{(R_1 + R_2)} = \frac{(180x20)}{(180+20)} = 18\Omega$$ ✓

$$V_1 = I_T R_{//} = 2x18 = 36V$$ ✓

$$V_2 = V_T - V_1 = 60 - 36 = 24V$$ ✓

$$R_x = \frac{V_2}{I_3} = \frac{24}{2} = 12\Omega$$ ✓ But $I_T = I_3 = I_1 + I_2 = 2A$ (4)

2.5.2 $R_T = R_{//} + R_x = 18 + 12 = 30\Omega$ ✓ (1)

2.5.3 $P = VI = 60x2 = 120W$ ✓ (1)
[18]

QUESTION 3

3.1
- Primary and secondary cells
- Relative movement of conductor and magnetic flux
- Difference of temperature between junctions of dissimilar metals (3)

3.2 Force which tends to produce an electric current in a circuit (2)

3.3
- Lines of magnetic flux never intersect.
- Each line of magnetic flux forms a closed loop.
- Lines of magnetic flux are like stretched elastic cords, always trying to shorten themselves.
- Lines of magnetic flux which are parallel and in the same direction repel one another.
- The direction of a line of magnetic flux at any point in a nonmagnetic medium such as air, is that of the north-seeking pole of a compass needle placed at that point. (5)

3.4 In loop ABEFA:

$$\Sigma E = \Sigma IR$$
$$28 - 20 = 0,5I_1 + 0,2(I_1 + I_2)$$
$$8 = 0,7I_1 + 0,2I_2 \text{.....(1)} \checkmark$$

In loop ACDFA:

$$\Sigma E = \Sigma IR$$
$$28 - 16 = 0,7I_2 + 0,2(I_1 + I_2)$$
$$12 = 0,2I_1 + 0,9I_2 \text{.....(2)} \checkmark$$

(2) x 3,5 :

$$42 = 0,7I_1 + 3,15I_2 \text{.....(3)} \checkmark$$

(3) – (1) :

$$34 = 2,95I_2$$
$$I_2 = 11,525A \checkmark$$

Subst. I_2 in (1):

$$8 = 0,7I_1 + 0,2I_2$$
$$8 = 0,7I_1 + 0,2(11,525) \checkmark$$
$$I_1 = 8,136A \checkmark$$

Therefore currents are:

$$I_1 = 8,136A \checkmark$$
$$I_2 = 11,525A \checkmark$$

(8)
[18]

QUESTION 4

4.1
- Interpoles
- Shifting the brushes
- Compensating windings
- Increasing brush contact resistance

(4)

4.2 Faraday's laws✓
- An emf induced in an electrical circuit whenever there is a change in the magnetic flux linking with that circuit.✓
- The magnitude of the induced emf is directly proportional to the rate of change of flux linkages.✓

(3)

4.3 4.3.1 The resistance of copper wire increases when the temperature of wire increases.

4.3.2 The resistance of carbon and insulators falls when a temperature rise occurs.

(2 × 2) (4)

4.4
- By increasing the current.
- By placing an iron core inside the solenoid.
- By increasing the number of turns keeping the current constant.

(Any 2 × 1) (2)

4.5 $I_L = \frac{P}{V} = \frac{10000}{440} = 22,7A$ ✓

$I_f = \frac{V}{R_f} = \frac{440}{161} = 2,7A$ ✓

$I_a = I_L - I_f = 22,7 - 2,7 = 20A$ ✓

$I_{start} = 1,5xI_a = 1,5x20 = 30A$ ✓

$V = E + I_a(R_a + R_{start})$ But $E_{start} = 0$ at standstill

$440 = 0 + 30(0,67 + R_{start})$

$R_{start} = 13,99\Omega$ ✓

(5)
[18]

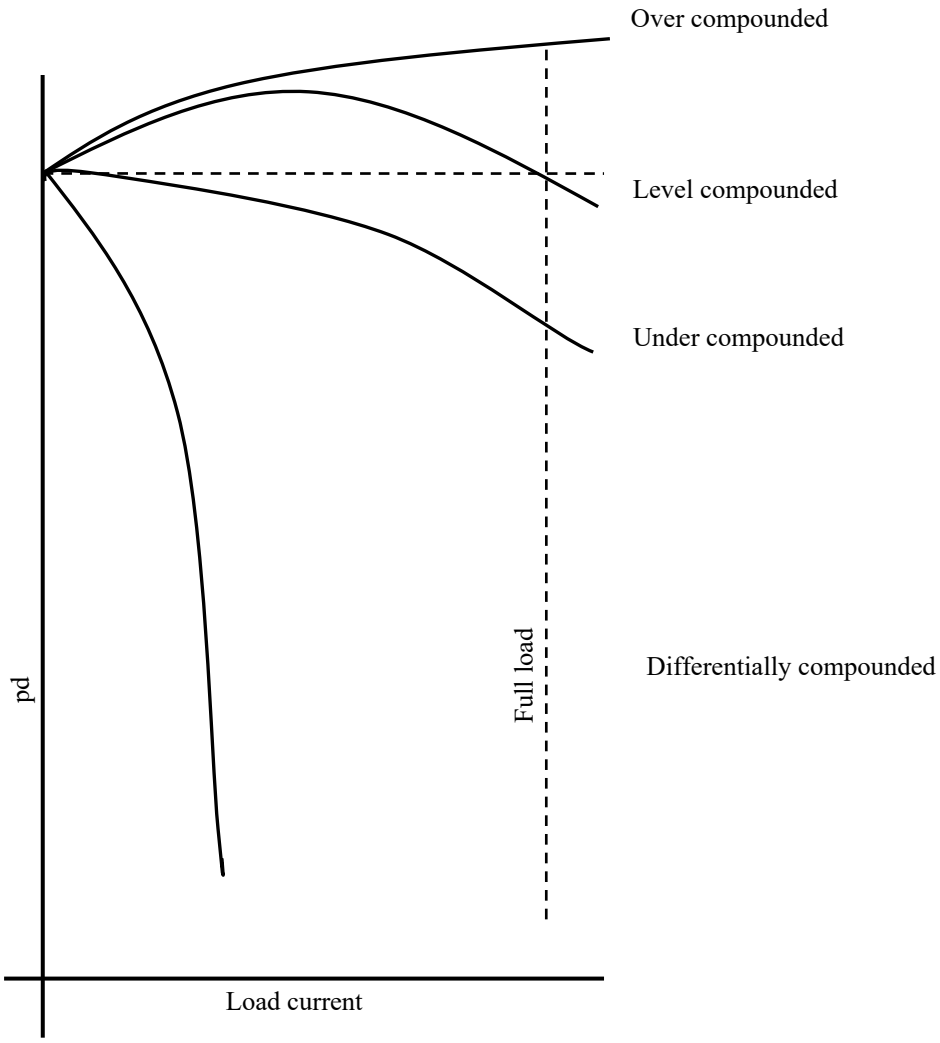
QUESTION 5

5.1
- Corrugations
- Auxiliary radiator tanks
- Round or elliptical section tubes
- Fins welded vertically to the tank sides

(4)

5.2 f = np
60 = n × 3
n = 20 r/s = 1 200 r/m ✓✓ (2)

5.3



(6)

5.4 $V_m = I_m R_m$

$= 0,3x75$ ✓

$= 22,5V$ ✓

$V_T = V_m + V_{se}$

$60 = 22,5 + V_{se}$ ✓

$V_{se} = 37,5V$ ✓

$R_{se} = \frac{V_{se}}{I_{se}} = \frac{V_{se}}{I_m}$

$= \frac{37,5}{0,3}$ ✓

$= 125\Omega$ ✓

(6)

[18]

QUESTION 6

6.1 It is used for protection of transformers from the faults occurring inside the transformer. (2)

6.2
- It provides a high starting torque.
- It causes a split phase to ensure self-starting
- It is also used in creation of phase angle difference to enable rotation.

(Any 2 × 1) (2)

6.3 They extract moisture from the air. (2)

6.4
- Ring feeders
- Radial feeders (2)

6.5 Radial feeder (1)

6.6
- Star connection
- Delta connection (2)

6.7
- Cheaper
- Less maintenance
- Less wear (brushes and commutator)
- Control circuits more reliable
- Supply (AC) always available
- Smaller and lighter
- Less maintenance than DC motors (Any 4 × 1) (4)

6.8
- Carbon emissions lead to global warming.
- Coal fired plants leave millions of tons of waste.
- Mining of coal destroys the face of the land.
- Displacement of humans as ground near mines may cave in. (Any 3 × 1) (3)

[18]

TOTAL: 100

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